

Look at This: Supervised Saliency Alignment for Visual Grounding in Vision-Language Models

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Abstract

Vision-language models (VLMs) learn rich image-text associations, yet they often attend to the wrong part of the image: the visual attention behind a generated word is noisy and tends to fall on background or image borders rather than the object the word names. We introduce the *Saliency Alignment Loss*, a simple auxiliary objective that teaches a VLM to look where it is describing. As the model generates a caption, we pull the attention behind each grounded word toward the image region that word refers to, supervised by the dense grounding annotations of COCONut-PanCap. The loss adds no parameters and no architectural changes, and drops into an ordinary fine-tuning run. Its effect is large and immediate: after only 200 steps on LLaVA-1.5-7B, the share of attention landing on the named object rises 4.6 $\times$, with matching gains on two complementary localization measures, while general question answering and hallucination behavior are left intact. We also pinpoint *where* the model changes: the language model, not the vision-language projector, absorbs the signal, and does so consistently across runs, concentrating the update in a small set of attention heads. Finally, we are clear about the method’s limits: this much sharper attention does *not*, on its own, raise downstream benchmark scores, evidence that attention localization is a capability worth supervising directly rather than a free path to better answers. We outline an instruction-tuned variant designed to turn these localization gains into downstream ones.

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1. Introduction

Vision-language models (VLMs) combine a pretrained vision encoder with a large language model and achieve strong performance across a broad range of multimodal tasks. Yet they remain unreliable on tasks that require fine-grained, spatially grounded understanding (counting, referential grounding, and spatial reasoning), where predictions often reflect shortcut correlations and scene-level priors rather than evidence drawn from the queried region (Chen et al., 2025; Sengupta et al., 2025).

A growing body of work locates part of this failure in how VLMs allocate visual attention. The attention from a generated token to the image patches can be read out as a *saliency map*, and the degree to which this map overlaps the relevant object correlates with answer correctness (Chen et al., 2025). In practice these maps are frequently mislocalized: attention disperses over non-informative regions, image borders, and background, and only a small subset of specialized heads attend in a grounding-relevant way (Kang et al., 2025b;a; Kim et al., 2025). Tellingly, *shaping* the attention distribution (suppressing distractor regions) helps more than simply increasing the attention mass on targets (Sengupta et al., 2025). Attention is thus both a faithful diagnostic of grounding and a promising intervention point.

Existing remedies stop short of supervising this signal directly. Inference-time methods reallocate or mask attention with hand-designed heuristics but never change what the model has learned. Representation-alignment objectives, which regularize a VLM’s hidden states toward a pretrained vision encoder (Yu et al., 2025; Yoon et al., 2025), do change the model, but their target is a *static*, query-agnostic feature map: it cannot express that a token should attend *here* for *this* prompt. What is missing is a training-time objective that supervises the token-conditioned attention itself against where the evidence actually is.

We close this gap with the **Saliency Alignment Loss**: an auxiliary objective that, during fine-tuning, pulls each generated token’s saliency map toward the image region its referent occupies. Using a densely grounded captioning corpus, COCONut-PanCap (Deng et al., 2025), we attach a target distribution (uniform over the token’s annotated

mask) to every grounded token and minimize the divergence from the model’s saliency through a Kullback–Leibler term added to the usual captioning loss. The objective introduces no new parameters and no architectural change. Because the goal is to change *where* the model attends, we measure that directly, on held-out images, with three complementary attention-localization metrics, before testing whether it transfers downstream.

Fine-tuning LLaVA-1.5-7B for only 200 steps already reshapes its attention: all three localization metrics improve sharply (mean Attention Mass Ratio $4.6\times$), and the maps become cleaner, concentrating on the objects a token names with markedly less spurious mass on background and image edges. We then ask *where* this adaptation lives, and find that the language model, not the multimodal projector, absorbs the gradient, reaching a reproducible, low-rank solution in a small, consistent set of attention heads, echoing the localization-head hypothesis (Kang et al., 2025a). We are equally clear about the method’s limits: these large gains in attention quality do *not* yet translate into downstream benchmark gains, so saliency-map quality and task grounding are for now dissociated, and alignment withdraws attention from broad background “stuff” (snow, sky) the annotations never cover. These observations motivate an in-progress variant that ports the objective into the standard LLaVA-1.5 instruction-tuning recipe to close the transfer gap.

Contributions.

- We propose the Saliency Alignment Loss, a parameter-free, architecture-agnostic training objective that supervises a VLM’s token-conditioned attention with region-level grounding annotations (Section 4).
- We define an intrinsic protocol for attention localization (AMR, AP, NSS) and show that alignment improves all three by large margins while preserving language-modeling quality (Sections 5 and 6).
- We give a mechanistic account of *where* the model adapts: the language model rather than the projector absorbs the signal, identically with or without the projector trainable, via a reproducible low-rank update to a few grounding-relevant heads; we also characterize a background-“stuff” failure mode and find that, for now, these localization gains do not transfer to downstream benchmarks (Section 7).

2. Related Work

Visual Attention Structure in VLMs. A growing body of research examines how vision-language models (VLMs) allocate visual attention in order to assess their visual grounding capabilities. Prior studies show that, although vision

tokens often receive limited overall attention, increasing attention mass alone does not improve spatial reasoning performance; instead, the *localization* of attention is the dominant factor (Chen et al., 2025). In particular, overlap between attention maps and ground-truth object regions strongly correlates with answer correctness, indicating that attention localization encodes meaningful grounding signals rather than serving as a purely interpretative tool.

Further work confirms that VLM attention is frequently mislocalized, with excessive focus on non-informative regions, degrading performance on visual grounding tasks such as counting and spatial reasoning (Kang et al., 2025b; An et al., 2025). Several strategies have been proposed to mitigate this issue, including attention reallocation and masking schemes. Notably, suppressing attention to distractor regions is often more effective than amplifying attention to target regions, highlighting the importance of shaping attention distributions rather than merely increasing their magnitude (Sengupta et al., 2025).

Finally, attention allocation varies substantially across layers and heads. Rather than being uniformly distributed, grounding-relevant attention is often concentrated in a small subset of specialized heads, which exhibit more accurate localization and improved interpretability (Kang et al., 2025a; Kim et al., 2025). These findings suggest that attention mechanisms constitute a promising intervention point for improving visual grounding in VLMs.

Supervised Representation Objectives. Several lines of work explore supervised objectives that align internal model representations with external visual signals. Representation alignment (REPA) has proven effective in other generative settings, for example diffusion models, where aligning hidden states with pretrained visual encoders improves both training efficiency and generation quality (Yu et al., 2025).

Building on this idea, recent work aligns internal representations of VLMs with those of pretrained vision encoders, leading to improved visual grounding and multimodal reasoning (Yoon et al., 2025). However, such approaches operate on static visual features that are independent of the textual prompt and generated tokens, limiting their ability to capture context-dependent grounding behavior.

More closely related to our work, Chefer et al. (2022) propose directly supervising attention maps in unimodal vision transformers using ground-truth region annotations, demonstrating improved robustness and domain generalization. Unlike static feature representations, attention maps in VLMs are inherently prompt- and token-dependent, evolving dynamically during text generation. This property enables context-sensitive visual grounding that cannot be captured by feature-level alignment alone. We build on this insight by extending attention-level supervision to the vision-

language setting, aligning token-conditioned attention maps with region-level grounding annotations during generation.

Existing approaches either apply inference-time attention heuristics, or align static visual representations that are independent of the generated text. To our knowledge, no prior work introduces a training-time attention supervision objective that explicitly aligns token-conditioned attention in VLMs with region-level grounding annotations. Our work fills this gap by supervising dynamic attention maps during text generation, enabling context-sensitive visual grounding.

3. Preliminaries

Vision-Language Models (VLMs). We consider VLMs that pair a Vision Transformer (ViT) (Dosovitskiy et al., 2021) with an autoregressive text decoder. The ViT encodes an input image $\mathbf{I}$ into a sequence of patch embeddings $\{v_i\}_{i=1}^N$, which serve as *patch tokens*. A projection module maps these tokens into the language model’s embedding space, where they act as conditioning context. The decoder then predicts tokens $\{w_t\}_{t=1}^T$ autoregressively under the standard next-token prediction objective. In LLaVA-1.5-7B, the ViT is a CLIP ViT-L/14 operating at 336×336 resolution, giving an $N = 24 \times 24 = 576$ patch grid, and the decoder is Vicuna-7B.

Saliency Maps. Because patch tokens and text tokens are processed jointly within the decoder’s self-attention layers, each generated token w_t attends to all N patch tokens. We form a saliency map $\mathbf{S}_t$ for w_t by aggregating these token-to-patch attention weights across layers and heads and reshaping the result onto the ViT patch grid, producing a spatial map that highlights the regions most relevant to predicting w_t . Throughout, we normalize $\mathbf{S}_t$ to a probability distribution over spatial locations so that maps are comparable across images and tokens. This saliency signal is the quantity our objective supervises (Section 4) and our metrics evaluate (Section 5).

4. Method

Our objective supervises *where* a VLM attends as it generates text, aligning each token’s saliency map with the image region its referent occupies. The idea is simple: when the model emits a word that names an annotated entity, the saliency for that word should land on that entity.

Token-level grounding signal. We assume training captions carry token-level grounding annotations. For a generated caption token c_t that refers to one or more image segments, let $\mathbf{M}_t \in \{0, 1\}^{H \times W}$ be the binary mask covering its annotated region(s). From the model’s attention we form the saliency map $\mathbf{S}_t$ over the N patch tokens (Section 3), bilinearly upsample it to the annotation resolution

$H \times W$, and apply a spatial softmax so that $\mathbf{S}_t$ is a distribution over pixels, $\sum_{ij} \mathbf{S}_{t,ij} = 1$. We supervise only the set $\mathcal{T}$ of *grounded* tokens (those annotated with at least one segment) and leave all other tokens (function words, punctuation) to the language modeling loss alone.

Alignment loss. We compare $\mathbf{S}_t$ against the target distribution that places probability uniformly over the annotated region,

$$\mathbf{T}_t = \mathbf{M}_t / |\mathbf{M}_t|, \quad (1)$$

and penalize their divergence. Our primary objective uses the Kullback–Leibler divergence,

$$\begin{aligned} \mathcal{L}_{\text{align}} &= \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \text{KL}(\mathbf{T}_t \parallel \mathbf{S}_t) \\ &= \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{i,j} \mathbf{T}_{t,ij} \log \frac{\mathbf{T}_{t,ij}}{\mathbf{S}_{t,ij}}. \end{aligned} \quad (2)$$

Because the target $\mathbf{T}_t$ is fixed, minimizing $\text{KL}(\mathbf{T}_t \parallel \mathbf{S}_t)$ is mass-covering: it drives $\mathbf{S}_t$ to put probability everywhere inside the mask and is heavily penalized wherever the mask is occupied but the saliency is near zero, encouraging the map to cover the *whole* referent rather than collapse onto a single patch. As a point of comparison we also implement a squared-error variant, $\mathcal{L}_{\text{align}}^{\text{MSE}} = \frac{1}{|\mathcal{T}|} \sum_t \sum_{ij} (\mathbf{S}_{t,ij} - \mathbf{T}_{t,ij})^2$, and use the KL form by default.

Training objective. We add the alignment term to the standard captioning cross-entropy,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{align}}, \quad (3)$$

where λ trades off fluency against grounding. The term introduces no new parameters and no architectural change, so it drops into a standard VLM fine-tuning pipeline. The vision encoder is kept frozen throughout; which of the remaining components actually absorbs the alignment gradient (the multimodal projector, the language model, or both) is a design choice we treat as an ablation (Section 5) and dissect mechanistically in Section 7.

5. Experimental Setup

5.1. Dataset and grounding signal

We draw our grounding signal from COCONut-PanCap (Deng et al., 2025), which pairs COCO images (Lin et al., 2015) with high-quality panoptic masks and human-verified, detailed captions. Crucially, each caption is *grounded*: mentioned entities are tagged with the panoptic segment(s) they refer to, in the form `<id: text>` (or `<id1, ..., idk: text>` for groups). We parse these tags to (i) recover the clean caption shown to the model and (ii) assign every caption token the segment id(s) it references; the per-token mask $\mathbf{M}_t$ of Section 4 is then the union of

the referenced segments in the panoptic mask. Relative to grounded-caption corpora such as GrandD (Rasheed et al., 2024) or PNG (González et al., 2021), COCONut-PanCap provides substantially denser annotations, yielding more supervised tokens per image. We hold out the last 10,000 annotated images for validation with a deterministic split and reuse it at evaluation, so all intrinsic metrics are computed on images the model never saw.

5.2. Training regimes

We study the objective under two recipes. The first is the focus of our analysis and is fully evaluated below; the second is an ongoing effort to convert intrinsic gains into downstream ones.

Direct fine-tuning. We fine-tune LLaVA-1.5-7B on grounded COCONut captions with the prompt “Describe the image in detail.” and the clean caption as target, using the objective of Equation (3) at $\lambda = 0.5$. The vision encoder is always frozen; since the term adds no parameters, the only structural choice is which component receives the gradient, and we compare the multimodal *projector* only, the *language model* only, and *both*. As a control we also run plain supervised fine-tuning with no alignment term ($\lambda = 0$), which isolates the effect of training on dense captions from the effect of the loss. Optimization uses AdamW (lr 2×10^{-5}), bf16-mixed precision, gradient checkpointing, a 3% warmup with cosine decay, and FSDP across 4 GH200 GPUs, for 200 steps, enough (as we show) to reshape attention. Saliency is extracted on the fly with an eager-attention backend.¹

Instruction tuning (in progress). Direct fine-tuning on dense captions risks over-optimizing the model toward captioning rather than general grounding. We therefore port the objective into the standard visual-instruction-tuning recipe, for two reasons: it is the industry-standard setup and so directly comparable to released models, and it lets us inject our caption-dense, grounded data *without* drowning the model in caption-heavy text. Concretely, starting from the LLaVA-1.5 stage-1 checkpoint (pretrained projector, not yet instruction-tuned), we instruction-tune on the standard LLaVA-1.5 665K instruction mixture, replacing only the detailed-description answers with grounded COCONut-PanCap captions; the rest of the mixture (conversation, reasoning, and academic VQA) is untouched, so the alignment loss applies to the grounded subset while the remainder preserves general instruction following. We use LoRA (Hu et al., 2022) adapters on the language model’s attention and MLP projections (with the projector trained directly), rather than full fine-tuning, to mitigate catastrophic forget-

ting of base capabilities: LoRA can only amplify directions already present in the model, which is well matched to our finding that the alignment adaptation is reproducible and concentrated in a stable set of attention heads (Appendix A). Results for this regime are forthcoming (Section 6.4).

5.3. Evaluation

We evaluate on two axes. Because the objective is designed to change *where* the model attends, we first measure attention localization directly, with three complementary *intrinsic* metrics on a held-out COCONut split we fully control; we then test whether this sharper attention carries over to a broad *downstream* suite. Some downstream benchmarks contain COCO images (Han et al., 2024), and so overlap our training data; we mark these with $\star$ in Table 2.

Intrinsic metrics. For each grounded token we measure how well its saliency localizes to its mask, using three complementary metrics, each computed per supervised token, averaged (ignoring undefined cases) over an image’s tokens, then over images. *Attention Mass Ratio (AMR)*, chance-normalized, is the fraction of saliency mass inside the mask divided by the chance level $|\mathbf{M}_t|/HW$:

$$AMR_t = \frac{\sum_{ij} \mathbf{S}_{t,ij} \mathbf{M}_{t,ij} / \sum_{ij} \mathbf{S}_{t,ij}}{|\mathbf{M}_t|/HW}, \quad (4)$$

so $AMR = 1$ is chance, >1 concentrates inside the mask; it is the test-time analog of our training target. *Average Precision (AP)* is the pixel-level area under the precision–recall curve with saliency as the score and mask membership as the label; unlike AMR it rewards correctly *ranking* in-mask pixels above out-of-mask ones. *Normalized Scanpath Saliency (NSS)* is the mean z -score of the saliency inside the mask after standardizing the map, giving the interpretable statement “saliency at annotated pixels is z standard deviations above the image-wide mean.”

Downstream suite. We report counting (CountBench (Paiss et al., 2023)), spatial reasoning (CV-Bench 2D/3D (Tong et al., 2024a)), general and fine-grained multimodal ability (MMStar (Chen et al., 2024), MMVP (Tong et al., 2024b)), low-level salient-object perception (the O3 split of Sal-Bench (Dahou et al., 2025)) and hallucination (POPE (Li et al., 2023)), and a blindness/bias probe (VLMs-are-Blind accuracy and a bias rate (Rahmanzadehgervi et al., 2024)). A per-configuration breakdown, including general VQA (GQA (Hudson & Manning, 2019)) and a second counting benchmark (TallyQA (Acharya et al., 2019), FSC-147 (Ranjan et al., 2021)), is deferred to Appendix B. We also track validation cross-entropy and next-token accuracy to confirm language modeling is preserved.

¹We release our saliency-extraction code as the vl-saliency package: <https://pypi.org/project/vl-saliency/>.

6. Results

6.1. Alignment sharpens attention localization

Table 1 reports the three localization metrics on held-out images, and the effect is large. The base model already places above-chance mass inside annotated regions ($\text{AMR} = 1.25$), confirming that grounding-relevant attention exists, but the alignment loss transforms it: training the language model raises mean AMR to 5.8, a $4.6\times$ increase and still $4\times$ on the outlier-robust median ($1.19 \rightarrow 4.70$), while AP more than doubles ($0.22 \rightarrow 0.55$) and NSS rises sevenfold ($0.26 \rightarrow 1.92$). All three move together: AMR confirms more mass lands in the mask, AP confirms in-mask pixels are genuinely ranked higher rather than the map being uniformly inflated, and NSS puts the contrast at nearly two standard deviations above the image mean: exactly the gains the objective targets, achieved in a short fine-tune without disturbing the model’s other behavior. Section 6.2 asks whether they also help downstream.

The ablation is unambiguous about *where* this happens. Plain caption fine-tuning without the loss (Caption SFT) barely moves the metrics ($\text{AMR} 1.46$), and aligning the projector alone is little better (1.51): the projector simply cannot reshape token-conditioned localization much. Aligning the language model is what matters, and adding the projector on top of the LM is essentially a no-op ($\text{LM} 5.76$ vs. $\text{LM+proj.} 5.90$, within their large spread). The two LM-trainable variants are statistically indistinguishable here, a behavioral equivalence we trace to the optimization geometry in Section 7.2. The high AMR variance ($\text{std} \approx 4.6$) reflects the metric’s sensitivity to small masks, which is why we report the median alongside; the median tells the same story.

6.2. The gains do not (yet) transfer downstream

A method that reshapes attention this dramatically should, one hopes, help on tasks that need grounding. It does not, at least not yet. Table 2 shows the LM-aligned models tracking the base model across the suite: small gains on spatial reasoning (CV-Bench, MMStar) are offset by small drops elsewhere (counting, MMVP, POPE), and no variant is consistently ahead. The per-benchmark differences among base, caption SFT, and the LM-aligned models are at the 1–2 point level, within run-to-run noise that we did not characterize with multiple seeds; we therefore read Table 2 for the *absence* of a consistent gain rather than for any single number. General VQA and hallucination are likewise preserved by LM-alignment (GQA and POPE; Appendix B). The projector-only model is the exception, and an instructive one: it is the only variant whose downstream profile changes sharply, lowering the bias rate ($41.7 \rightarrow 35.9$) but degrading hallucination (POPE $84.3 \rightarrow 73.2$) and counting, a capability-for-behavior trade we explain in Section 7.1.

This gap between large intrinsic gains and flat downstream numbers is the central tension of the direct-fine-tuning regime, and it has two plausible causes: training on dense captions may push the model toward a captioning style that the benchmarks do not reward, and sharper attention may simply not be what bottlenecks these tasks for a 7B model. The instruction-tuning regime of Section 5.2 targets the first cause directly; we return to it in Section 6.4.

6.3. Qualitative effect on saliency

Figure 1 makes the metrics tangible through two representative cases; it is meant to illustrate the effect, not to quantify it (the systematic version is Table 1). First, alignment *cleans up* the maps: the base model frequently deposits bright spots on non-informative regions (image edges and, characteristically, the bottom border) and distributes mass unevenly across the objects a token names, whereas after alignment these spurious peaks are suppressed and the mass redistributes onto the referent more uniformly (the flying birds), exactly the mass-covering behavior Equation (2) induces.

Second, the same pressure that concentrates attention on discrete objects can *over-concentrate* at the expense of broad backgrounds: for a token like `snow`, the aligned model withdraws the saliency the base model sensibly spread across the snowy ground, whereas background *objects* with definite extent (plants, branches) stay well localized. We take up this asymmetry, and its root in the annotation distribution, in Section 7.3.

6.4. Toward downstream transfer: instruction tuning

The instruction-tuned regime of Section 5.2 targets the first cause directly: porting the objective into the standard LLaVA-1.5 665K mixture via LoRA lets the grounded captions enter as one part of a diverse instruction diet rather than the entire training signal, aiming to retain the localization gains of Table 1 without over-fitting to a captioning style. These experiments are ongoing.

7. Analysis

Having shown that alignment reshapes attention intrinsically (Table 1) without moving downstream numbers (Table 2), we now ask what the three configurations actually do, where in the model they do it, and where the method breaks.

7.1. A behavioral dissociation between components

The three adaptation sites do not produce one uniform “improvement from saliency supervision”; they produce distinct phenotypes. The LM-aligned models (LM-only and LM+projector) earn by far the cleanest saliency (Table 1) yet sit at base-model level downstream (Table 2), with capa-

Table 1. Intrinsic attention-localization on the held-out COCONut-PanCap split (Section 5.3); each cell is mean / median / std over images. Higher is better; AMR = 1 is chance. All aligned variants use the KL objective at $\lambda = 0.5$ for 200 steps and differ only in the trainable component; *Caption SFT* is the no-alignment control ($\lambda = 0$). Best mean per metric in **bold**.

Variant	AMR (mean/med/std)	AP (mean/med/std)	NSS (mean/med/std)
LLaVA-1.5-7B (base)	1.25 / 1.19 / 0.31	0.22 / 0.19 / 0.12	0.26 / 0.18 / 0.33
Caption SFT ($\lambda=0$, LM+proj.)	1.46 / 1.36 / 0.43	0.24 / 0.21 / 0.13	0.42 / 0.34 / 0.39
+ KL align, projector only	1.51 / 1.40 / 0.45	0.27 / 0.24 / 0.14	0.72 / 0.63 / 0.49
+ KL align, LM only	5.76 / 4.70 / 4.60	0.55 / 0.54 / 0.18	1.92 / 1.81 / 0.84
+ KL align, LM + projector	5.90 / 4.80 / 4.78	0.56 / 0.55 / 0.18	1.94 / 1.84 / 0.85

Table 2. Downstream benchmarks (%); higher is better except bias rate ($\downarrow$). Best per column in **bold**. Models as in Table 1. Despite the large intrinsic gains, the aligned models track the base model; projector-only trades grounding (POPE, MMVP) for a lower bias rate.

* contains COCO images (Han et al., 2024), which overlap our training data.

Variant	Count	CV-2D*	CV-3D	MMStar	MMVP	O3	POPE*	VLMsB	bias $\downarrow$
LLaVA-1.5-7B (base)	40.7	55.8	52.2	33.1	62.0	44.7	84.3	16.2	41.7
Caption SFT ($\lambda=0$)	38.9	56.7	53.2	35.1	61.0	43.7	83.7	15.6	42.1
KL align, projector only	33.0	49.9	53.8	32.1	50.7	40.3	73.2	16.5	35.9
KL align, LM only	39.9	57.9	53.8	35.2	59.7	43.9	83.5	15.8	38.9
KL align, LM + projector	39.5	57.6	53.4	35.3	57.7	44.2	83.6	15.4	40.3

bilities preserved. The projector-only model is the opposite: its saliency barely improves over plain caption fine-tuning, but it is the one variant whose downstream behavior shifts sharply: hallucination worsens (POPE 84.3 $\rightarrow$ 73.2) and the bias rate falls (41.7 $\rightarrow$ 35.9).

The takeaway is a clean *decoupling*: saliency-map quality does not predict downstream grounding. Counting makes this sharpest. Across components, attention quality and counting accuracy are not merely uncorrelated but *anti*-correlated: training the projector *improves* TallyQA counting while training the language model, which yields by far the best maps, degrades it (Appendix B). The model that looks best does not answer best. This echoes the attention-overlap-versus-counting null we find across attribution methods (Appendix D), and it cautions against treating sharper attention as a proxy for better grounding.

7.2. Where the adaptation lives

A parameter-drift analysis (Appendix A) explains why the LM-aligned variants behave identically: the language model, not the projector, absorbs the alignment signal and reaches essentially the same solution whether or not the projector is also trained. The update is reproducible and concentrated in a small set of attention heads, consistent with the localization-head hypothesis (Kang et al., 2025a; Kim et al., 2025) under explicit supervision, and stable enough to motivate the LoRA-based regime of Section 5.2. The projector-only model is the exception: its large projector drift feeds the frozen language model visual features off their training distribution, the likely source of the POPE/grounding

degradation noted above.

7.3. A failure mode on background “stuff”

The one place alignment clearly hurts is broad, amorphous background: “stuff” classes such as snow and sky (the snow case of Section 6.3). After fine-tuning, the base model’s sensible spreading of saliency across a snowy ground is replaced by a withdrawal of mass onto foreground objects. We attribute this to the *supervision distribution* rather than the objective itself. Grounded captions reference discrete, nameable entities (individual objects and groups) and almost never attach a token to a large background region, so the alignment target $\mathbf{T}_t$ practically never places mass on stuff; because cross-entropy still rewards *mentioning* the background while the alignment term only ever pulls mass toward objects, the model learns to treat “unannotated” as “unattended.” The control is telling: background entities that *are* annotated because they have definite extent (plants, branches) stay well localized: it is specifically the un-annotatable, region-filling categories that degrade. The fix follows directly: COCONut provides full *panoptic* masks including stuff segments, so extending supervision to stuff tokens (or simply exempting background tokens from the alignment term) would remove the implicit “attend only to objects” pressure. We leave stuff-aware supervision to future work.

8. Limitations

We frame these results as a focused study of a single objective, with a few limitations worth stating.



Figure 1. Saliency for an object token before (left) and after (right) alignment; brighter is more attention (Section 6.3). Alignment cleans up the map: attention the base model scattered onto background and image borders is pulled onto the birds, and the per-object mass is more evenly distributed. We visualize the token rows because LLaVA tokenizes “crows” as `_c+rows`, so the grounded subword is `rows`.

No head-to-head baselines. We do not empirically compare against the methods we position against (feature-alignment objectives such as REPA/VIRAL (Yu et al., 2025; Yoon et al., 2025), inference-time attention steering, or attention supervision in unimodal ViTs (Chefer et al., 2022)); our claims concern the effect of the objective, not state-of-the-art performance.

Intrinsic metrics measure localization, not downstream value. AMR, AP, and NSS quantify how well attention falls on the annotated region, which is exactly what the loss optimizes (AMR especially is close to the training target); the downstream suite, not these metrics, is what tests usefulness.

Limited scope. Results are for one architecture (LLaVA-1.5-7B), one loss weight ($\lambda=0.5$), 200 steps, and a single seed; the downstream deltas in Table 2 are within run-to-run noise, and a λ sweep, longer schedules, and the squared-error variant remain untested.

Mechanistic claims are correlational. The drift analysis (Appendix A) identifies candidate localization heads by how much their weights move, not by causal ablation, and reproducibility is shown across configurations rather than seeds.

9. Conclusion

We introduced the Saliency Alignment Loss, a parameter-free, architecture-agnostic auxiliary objective that super-

vises a VLM’s token-conditioned attention with region-level grounding annotations. On LLaVA-1.5-7B, only 200 steps on grounded COCONut-PanCap captions transform attention localization (mean Attention Mass Ratio rises $4.6\times$, Average Precision doubles, and Normalized Scanpath Saliency rises sevenfold) while leaving language modeling intact, and the qualitative maps confirm it: mass the base model scattered onto image borders and background is redrawn onto the objects a token names.

Our analysis pinpoints *where* this happens: the language model, not the multimodal projector, absorbs the signal, identically whether or not the projector is trainable, reaching a reproducible solution concentrated in a small set of attention heads (Appendix A). This echoes the localization-head hypothesis, now induced by explicit supervision, and helps explain the method’s data efficiency.

We are also clear about what the present recipe does *not* yet do: the large gains in attention quality do not translate into higher downstream benchmark scores, so saliency-map quality and task grounding are, for now, dissociated. We read this not as a defect of the objective but as evidence that localization is a capability worth supervising in its own right. We also identified a clean failure mode (alignment withdraws attention from background “stuff” the annotations never cover) and its direct fix in stuff-aware supervision.

These findings set up the work in progress: porting the objective into the standard LLaVA-1.5 instruction-tuning recipe with LoRA, to retain the intrinsic gains while preserv-

ing general capabilities and to test whether decoupling the signal from a caption-only diet closes the transfer gap. We hope that treating attention as a directly supervisable signal, rather than only a diagnostic one, proves a useful lever for building VLMs that look where they should.

References

- Acharya, M., Kafle, K., and Kanan, C. Tallyqa: Answering complex counting questions. In *AAAI*, 2019.
- An, W., Tian, F., Leng, S., Nie, J., Lin, H., Wang, Q., Dai, G., Chen, P., and Lu, S. Mitigating object hallucinations in large vision-language models with assembly of global and local attention. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 29915–29926, June 2025.
- Chefer, H., Schwartz, I., and Wolf, L. Optimizing relevance maps of vision transformers improves robustness. In *Advances in Neural Information Processing Systems*, 2022.
- Chen, L., Li, J., Dong, X., Zhang, P., Zang, Y., Chen, Z., Duan, H., Wang, J., Qiao, Y., Lin, D., and Zhao, F. Are we on the right way for evaluating large vision-language models?, 2024. URL <https://arxiv.org/abs/2403.20330>.
- Chen, S., Zhu, T., Zhou, R., Zhang, J., Gao, S., Niebles, J. C., Geva, M., He, J., Wu, J., and Li, M. Why is spatial reasoning hard for vlms? an attention mechanism perspective on focus areas. In *Forty-second International Conference on Machine Learning*, 2025.
- Dahou, Y., Huynh, N. D., Le-Khac, P. H., Para, W. R., Singh, A., and Narayan, S. Vision-language models can’t see the obvious, 2025. URL <https://arxiv.org/abs/2507.04741>.
- Deitke, M., Clark, C., Lee, S., Tripathi, R., Yang, Y., Park, J. S., Salehi, M., Muennighoff, N., Lo, K., Soldaini, L., Lu, J., Anderson, T., Bransom, E., Ehsani, K., Ngo, H., Chen, Y., Patel, A., Yatskar, M., Callison-Burch, C., Head, A., Hendrix, R., Bastani, F., VanderBilt, E., Lambert, N., Chou, Y., Chheda, A., Sparks, J., Skjonsberg, S., Schmitz, M., Sarnat, A., Bischoff, B., Walsh, P., Newell, C., Wolters, P., Gupta, T., Zeng, K.-H., Borchardt, J., Groeneveld, D., Nam, C., Lebrecht, S., Wittliff, C., Schoenick, C., Michel, O., Krishna, R., Weihs, L., Smith, N. A., Hajishirzi, H., Girshick, R., Farhadi, A., and Kembhavi, A. Molmo and pixmo: Open weights and open data for state-of-the-art vision-language models, 2024. URL <https://arxiv.org/abs/2409.17146>.
- Deng, X., Yu, Q., Athar, A., Yang, C., Yang, L., Jin, X., Shen, X., and Chen, L.-C. Coconut-pancap: Joint panoptic segmentation and grounded captions for fine-grained understanding and generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*, 2021.
- González, C., Ayobi, N., Hernández, I., Hernández, J., Pont-Tuset, J., and Arbeláez, P. Panoptic narrative grounding, 2021. URL <https://arxiv.org/abs/2109.04988>.
- Han, X., Wang, Y., Zhai, B., You, Q., and Yang, H. Coco is “all” you need for visual instruction fine-tuning, 2024. URL <https://arxiv.org/abs/2401.08968>.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://arxiv.org/abs/2106.09685>.
- Hudson, D. A. and Manning, C. D. Gqa: A new dataset for real-world visual reasoning and compositional question answering, 2019. URL <https://arxiv.org/abs/1902.09506>.
- Kang, S., Kim, J., Kim, J., and Hwang, S. J. Your large vision-language model only needs a few attention heads for visual grounding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 9339–9350, 2025a.
- Kang, S., Kim, J., Kim, J., and Hwang, S. J. See what you are told: Visual attention sink in large multimodal models. In *The Thirteenth International Conference on Learning Representations*, 2025b.
- Kim, J., Kang, S., Park, J., Kim, J., and Hwang, S. J. Interpreting attention heads for image-to-text information flow in large vision-language models. In *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025.
- Leem, S. and Seo, H. Attention guided cam: visual explanations of vision transformer guided by self-attention. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pp. 2956–2964, 2024.
- Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, W. X., and Wen, J.-R. Evaluating object hallucination in large vision-language models, 2023. URL <https://arxiv.org/abs/2305.10355>.
- Lin, T.-Y., Maire, M., Belongie, S., Bourdev, L., Girshick, R., Hays, J., Perona, P., Ramanan, D., Zitnick, C. L.,

- and Dollár, P. Microsoft coco: Common objects in context, 2015. URL <https://arxiv.org/abs/1405.0312>.
- Paiss, R., Ephrat, A., Tov, O., Zada, S., Mosseri, I., Irani, M., and Dekel, T. Teaching clip to count to ten. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 3147–3157. IEEE, 2023.
- Rahmanzadehgervi, P., Bolton, L., Taesiri, M. R., and Nguyen, A. T. Vision language models are blind, 2024. URL <https://arxiv.org/abs/2407.06581>.
- Ranjan, V., Sharma, U., Nguyen, T., and Hoai, M. Learning to count everything, 2021. URL <https://arxiv.org/abs/2104.08391>.
- Rasheed, H., Maaz, M., Mullappilly, S. S., Shaker, A., Khan, S., Cholakal, H., Anwer, R. M., Xing, E., Yang, M.-H., and Khan, F. S. Glamm: Pixel grounding large multimodal model, 2024. URL <https://arxiv.org/abs/2311.03356>.
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. Grad-cam: Visual explanations from deep networks via gradient-based localization. *International Journal of Computer Vision*, 128(2):336–359, October 2019. ISSN 1573-1405. doi: 10.1007/s11263-019-01228-7. URL <http://dx.doi.org/10.1007/s11263-019-01228-7>.
- Sengupta, S., Moradinasab, N., Liu, J., and Brown, D. E. Can vision-language models count? a synthetic benchmark and analysis of attention-based interventions. *arXiv preprint arXiv:2511.17722*, 2025.
- Tong, S., Brown, E., Wu, P., Woo, S., Middepogu, M., Akula, S. C., Yang, J., Yang, S., Iyer, A., Pan, X., Wang, Z., Fergus, R., LeCun, Y., and Xie, S. Cambrian-1: A fully open, vision-centric exploration of multimodal llms, 2024a. URL <https://arxiv.org/abs/2406.16860>.
- Tong, S., Liu, Z., Zhai, Y., Ma, Y., LeCun, Y., and Xie, S. Eyes wide shut? exploring the visual shortcomings of multimodal llms, 2024b. URL <https://arxiv.org/abs/2401.06209>.
- Yoon, H., Jung, J., Kim, J., Choi, H., Shin, H., Lim, S., An, H., Kim, C., Han, J., Kim, D., et al. Visual representation alignment for multimodal large language models. *arXiv preprint arXiv:2509.07979*, 2025.
- Yu, S., Kwak, S., Jang, H., Jeong, J., Huang, J., Shin, J., and Xie, S. Representation alignment for generation: Training diffusion transformers is easier than you think. In *The Thirteenth International Conference on Learning Representations*, 2025.

A. Where the Adaptation Lives: Parameter-Drift Analysis

This appendix gives the parameter-drift analysis summarized in Section 7.2: it measures how far each component moves from the base model and shows that the language model does the work, reproducibly.

The language model absorbs the signal. The language model’s relative update is essentially *invariant* to whether the projector is also trainable (7.72×10^{-3} for LM-only versus 7.67×10^{-3} for LM+projector, agreeing to three significant figures), while under joint training the projector moves only 22% of its solo magnitude (1.36×10^{-2} vs. 6.07×10^{-2}). The loss splits its gradient across components, but the language-model solution dominates and the two do not compose: LM+projector is the LM solution, not a superposition. The projector-only model’s large projector drift, in turn, feeds the frozen LM visual features off its training distribution, the likely source of its POPE/grounding degradation (Section 7.1). The vision tower is confirmed frozen, its drift at conversion-noise level across all runs.

The update is reproducible and head-concentrated. Within the language model the update is distributed across all decoder layers but modestly concentrated (layer-level Gini ≈ 0.10 – 0.13 for attention projections and ≈ 0.06 for MLP) and sharper at the head level (Gini up to 0.29), including a standout head (layer 11 query projection, head 22) that dominates its layer. Critically, this structure is *reproducible*: between the LM-only and LM+projector runs the top-drifting layers overlap with Jaccard = 1.00, and the top heads with Jaccard = 1.00 in 8 of 10 analyzed cells. The optimizer reaches the same low-dimensional attractor regardless of the projector, consistent with the localization-head phenomenon (Kang et al., 2025a; Kim et al., 2025), here arising from *explicit* spatial supervision rather than unsupervised training. This reproducible, low-rank signature is what makes the objective a natural fit for the LoRA-based instruction-tuning regime of Section 5.2. We caution that drift is a proxy for functional change, and that we have verified reproducibility across configurations but not yet across random seeds.

B. Per-Configuration Benchmark Breakdown

This appendix expands the consolidated downstream results (Table 2) into the per-component, per-loss breakdown from our evaluation round, and adds two measurements the main table omits: general visual question answering (GQA) and a second counting benchmark (TallyQA, FSC-147). All numbers are zero-shot.

Capability preservation. Table 3 reports GQA and POPE. The picture sharpens the dissociation of Section 7.1: aligning the *language model* (alone or with the projector) leaves general VQA and hallucination essentially untouched (POPE F1 even ticks up), whereas aligning the *projector alone* degrades both markedly (GQA $0.606 \rightarrow 0.580$, POPE F1 $0.839 \rightarrow 0.721$). This is consistent with the drift analysis (Section 7.2): the projector-only run moves the projector hardest, feeding the frozen language model visual features that have drifted off its training distribution.

Table 3. General VQA and hallucination (zero-shot). Higher is better; best per column in **bold**. Aligning the language model preserves capability; aligning the projector alone degrades it. * GQA and POPE both draw on COCO images.

Variant	GQA acc.*	POPE acc.*	POPE F1*
LLaVA-1.5-7B (base)	0.606	0.854	0.839
Caption SFT ($\lambda=0$)	0.602	0.857	0.844
KL align, projector only	0.580	0.778	0.721
KL align, LM only	0.596	0.859	0.846
KL align, LM + projector	0.599	0.852	0.836

Counting. Table 4 reports TallyQA accuracy as a function of which component is trained and whether the alignment loss is on. The pattern is striking: training the *projector* improves counting (even without the loss, $40.96 \rightarrow 42.67$; with it, 45.06), while training the *language model* degrades it ($\rightarrow 35.82$ without the loss, 34.38 with), and the joint run follows the language model down. The alignment loss does not create these effects so much as *amplify* each component’s native direction.

Crucially, language-model alignment is exactly the configuration that produces the largest saliency gains (Table 1) yet the worst counting, so across components, saliency-map quality and counting accuracy are not merely uncorrelated but *anti*-correlated. This is the same message as the attention-overlap-vs-counting null of Appendix D, now seen across training configurations. (Counting benchmarks themselves disagree on the projector, which is best here on TallyQA but worst on

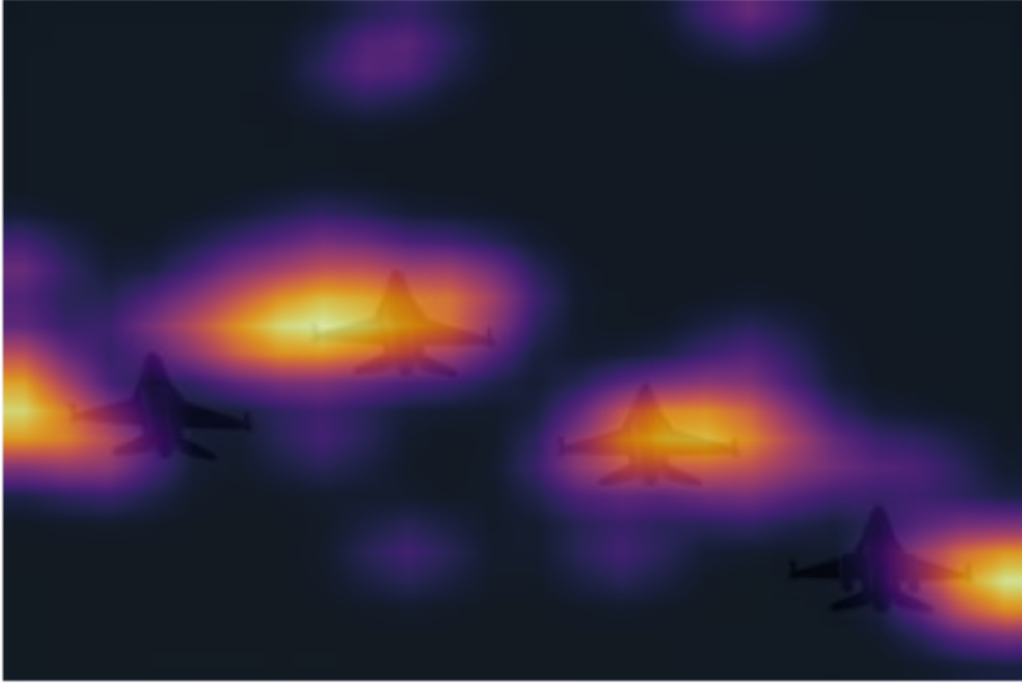


Figure 2. Attention saliency when the model is asked to count four planes.

CountBench in Table 2; the robust signal is that language-model alignment never improves counting despite producing the best maps.) FSC-147 is near chance for every variant ($\approx 5\text{--}6\%$ accuracy, with diverging mean error), so we omit it from the table; the models simply cannot do open-set counting at these magnitudes.

Table 4. TallyQA (which includes COCO images) counting accuracy (%) by trained component and loss. Base (no fine-tuning) scores 40.96. Training the projector helps counting; training the language model hurts it; the loss amplifies the component’s direction.

Trained component	SFT ($\lambda=0$)	+ KL align ($\lambda=0.5$)
Projector	42.67	45.06
Language model	35.82	34.38
LM + projector	37.25	33.57

C. Reading Saliency out of a VLM

Before settling on attention as the signal to supervise, we surveyed how to read saliency out of a VLM at all. This appendix documents that exploratory study; it motivates the design choices of Section 3 (mean aggregation over layers and heads; attention rather than gradients) and is not part of the main evaluation.

Saliency maps visualize the local importance of image regions for predicting a token. They are most commonly derived from the VLM’s attention, though some methods also incorporate the prediction’s gradients. We investigated raw attention, raw gradients, GradCAM (Selvaraju et al., 2019), and AGCAM (Leem & Seo, 2024), computing gradients from the backward pass of the token of interest (Figure 3 compares all four). Map quality varied strongly with both the image and the prompt.

Raw attention was typically sparse. Sometimes it gave a very clear read on where the model looked (Figure 2); other times it scattered seemingly at random onto background or image edges. Unlike gradient methods, attention placed high scores on only a few patch tokens (e.g., one or two patches near a cat’s head), which covers small objects well but leaves large ones only partially highlighted.

Raw gradients behaved oppositely: broadly distributed, with high scores on many tokens. In useful cases two regimes

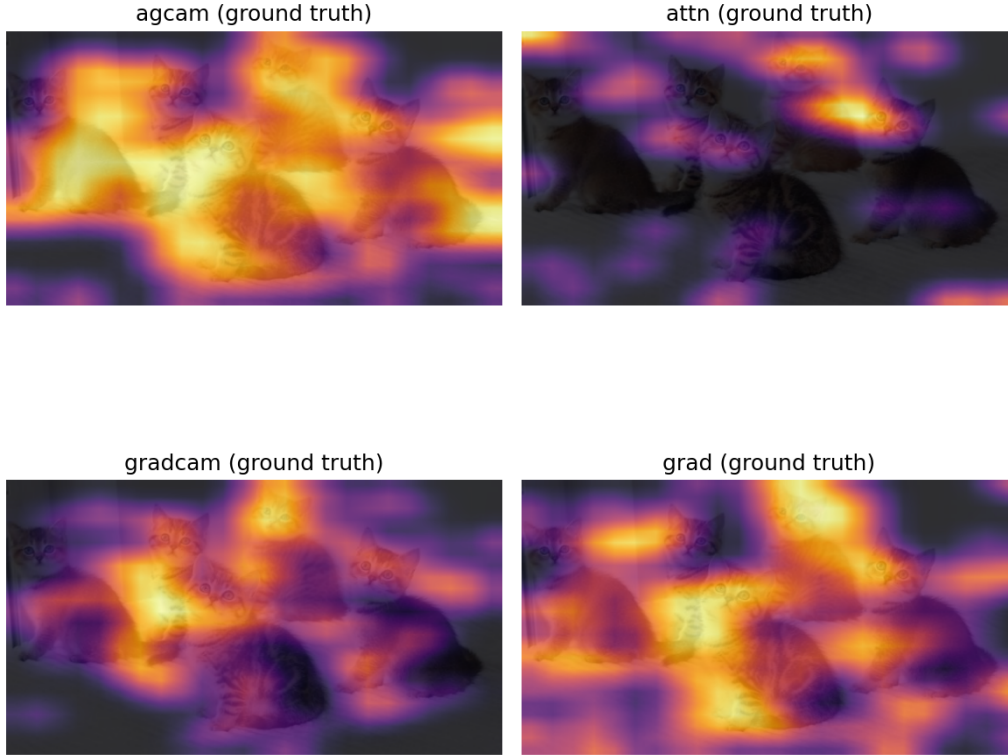


Figure 3. The four attribution methods for the token “four” in the prompt “How many cats are in the image? There are four cats.”

appeared: either the gradients fully covered the region of interest, or they concentrated entirely on irrelevant areas, the inverse of the correct map. Our maps suggest, though do not confirm, that the first regime coincides with correct predictions and the second with incorrect ones. We also observed no visual correspondence between a layer’s attention map and its gradient map.

CAM variants mix attention and gradients with simple transforms (e.g., ReLU). These were hit-or-miss: when they worked they produced maps that cleanly covered the objects of interest, but roughly a quarter of the time they were less interpretable than either ingredient. AGCAM (which additionally applies a sigmoid to the attention) produced the best maps when it worked but was less consistent than the more reliably interpretable GradCAM. We see room for a CAM variant tuned specifically to VLMs.

Layers and heads. Early-layer attention maps were far more interpretable than late-layer ones, where contextualized tokens attend less to the image. We saw strong evidence for localization heads (Kang et al., 2025a) (a few heads do most of the looking and yield the cleanest maps), though the spatial-entropy heuristic for finding them was only mediocre.

Aggregation and transforms. Among aggregation rules, mean, max, and sum worked best, in that order; individual heads were complementary, each attending where others did not. Binarization and Gaussian smoothing both improved interpretability.

D. Saliency and Counting

Counting motivated much of this work, and we report here two negative results that shaped our focus on direct attention supervision.

Counting is deceptively hard for VLMs: they must map visual features to a discrete numeral with no explicit enumeration mechanism. We observe that saliency maps localize cleanly to the relevant objects only when the count is small (typically fewer than five), suggesting that current models lean on holistic pattern recognition rather than individuated object identification as counts grow.

Count-specific fine-tuning does not sharpen saliency. Contrary to results on unimodal vision transformers, where count supervision sharpens attention toward the relevant regions (Paiss et al., 2023), we find this does not transfer to VLMs. Fine-tuning LLaVA-1.5-7B and Gemma 3 4B on PixMo-Count (Deitke et al., 2024) improved counting accuracy by roughly 10% but left saliency localization unchanged: accuracy and grounding moved independently.

Saliency overlap does not predict counting accuracy. We tested whether a model’s internal spatial focus tracks its counting accuracy, as it would if counting were grounded rather than purely linguistic. Extracting saliency under several attribution methods and measuring its overlap with ground-truth object masks on a COCO-derived subset of TallyQA (Acharya et al., 2019), we found that saliency mass inside the annotated regions does *not* correlate with correct counts.

Together these results say that counting accuracy and attention localization are decoupled under the standard objective: improving the answer does not improve the looking, and *vice versa*. This is precisely what motivates supervising localization directly, as in the main paper, rather than hoping it emerges from a task loss.

E. Saliency Hallucination

A final observation concerns how saliency behaves for *incorrect* predictions. When a VLM miscounts (answering “four” for an image of three cats), we often (though not always) see a phantom blob of attention in the background, structurally similar to the mass on the three real objects, as if the model attends to evidence for an object that is not there. The complementary error also occurs: when the model undercounts (“two” for three cats), the saliency skips the missed object entirely. The effect was visible in somewhat more than half the images we examined, most clearly for large objects and small counts. We report it as a qualitative phenomenon; quantifying this “saliency hallucination” and relating it to answer errors is an interesting direction we did not pursue here.